



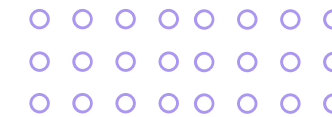
DEPLOYING AND MANAGING A CLOUD ENVIRONMENT USING OPENSTACK

Presented By : Azzman Nor Elimane
Elghazi Silia



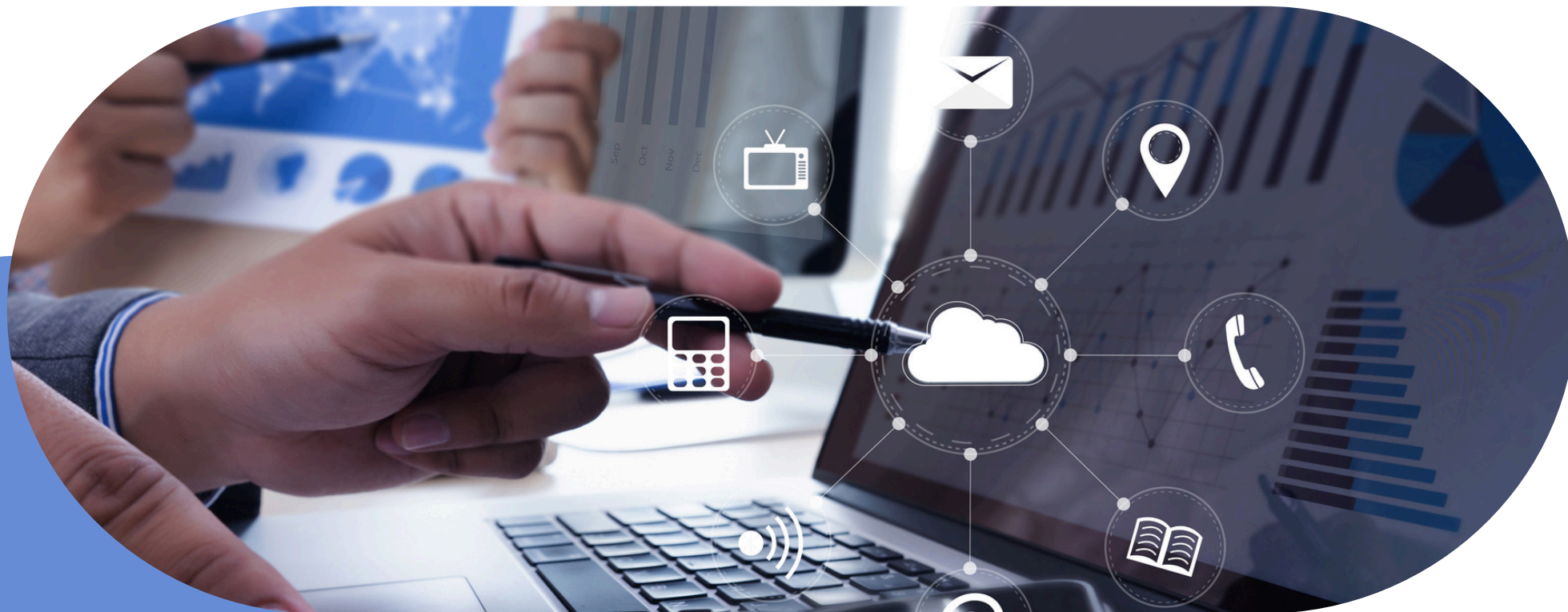
TABLE OF CONTENT

- 01 INTRODUCTION
- 02 FUNDAMENTALS OF CLOUD COMPUTING
- 03 OPENSTACK ARCHITECTURE
- 04 INSTALLATION AND CONFIGURATION
- 05 VIRTUAL MACHINE DEPLOYMENT
- 06 STORAGE AND NETWORKING SETUP
- 07 CHALLENGES AND TROUBLESHOOTING



INTRODUCTION

- **Objective:** Deploy a functional cloud infrastructure using OpenStack
- **Approach:** Step-by-step deployment from image to web server
- **Tools:** MicroStack, Horizon, Nova, Neutron...

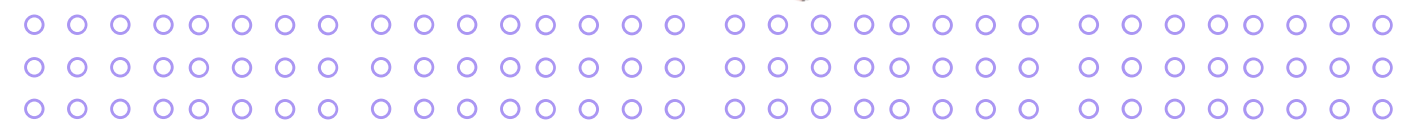
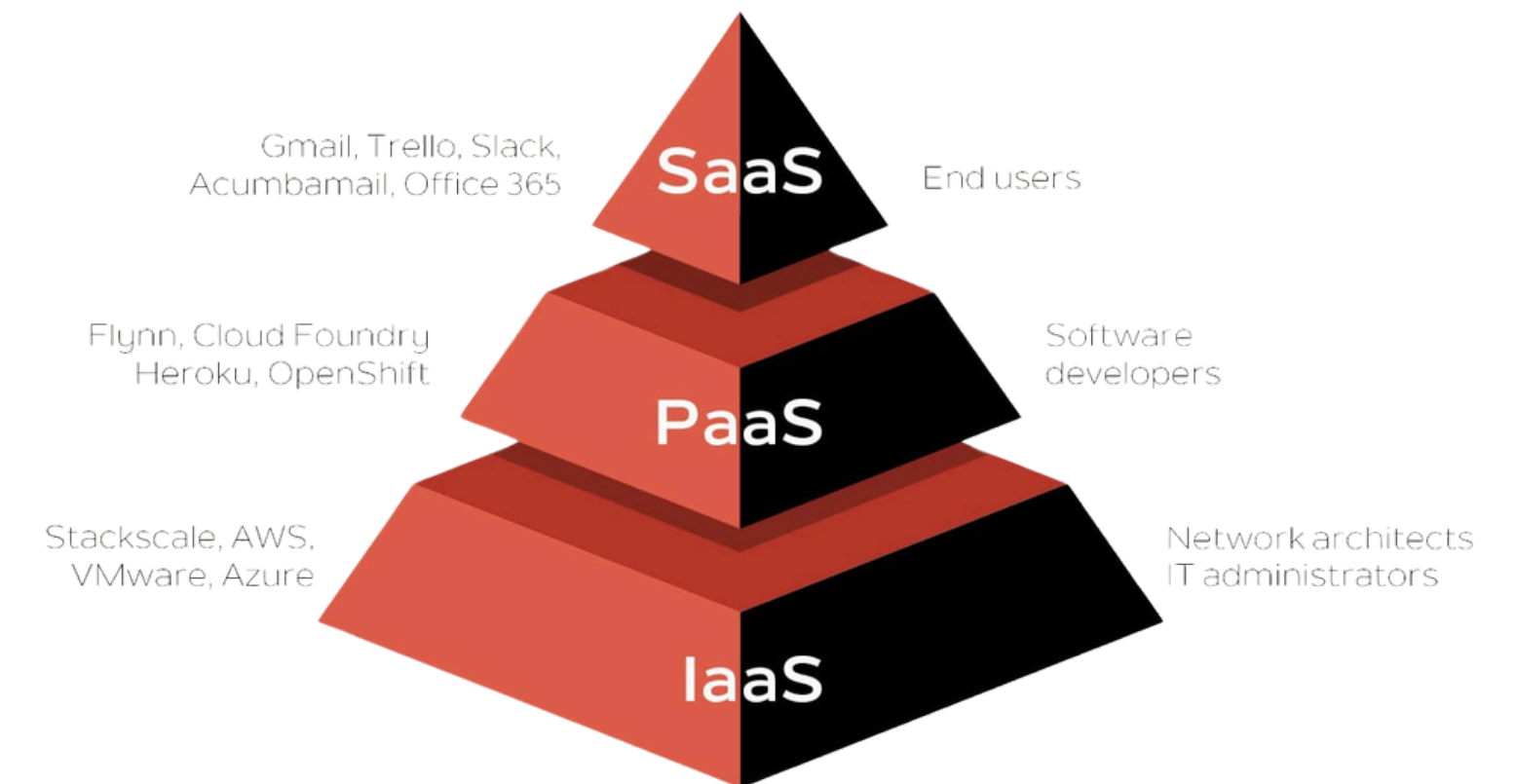


FUNDAMENTALS OF CLOUD COMPUTING

- Cloud models: IaaS, PaaS, SaaS
- Cloud types: Public, Private, Hybrid
- Benefits: Scalability, Elasticity, On-demand Resources



Cloud service models

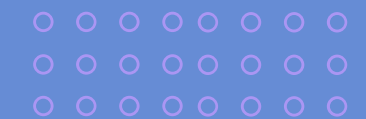
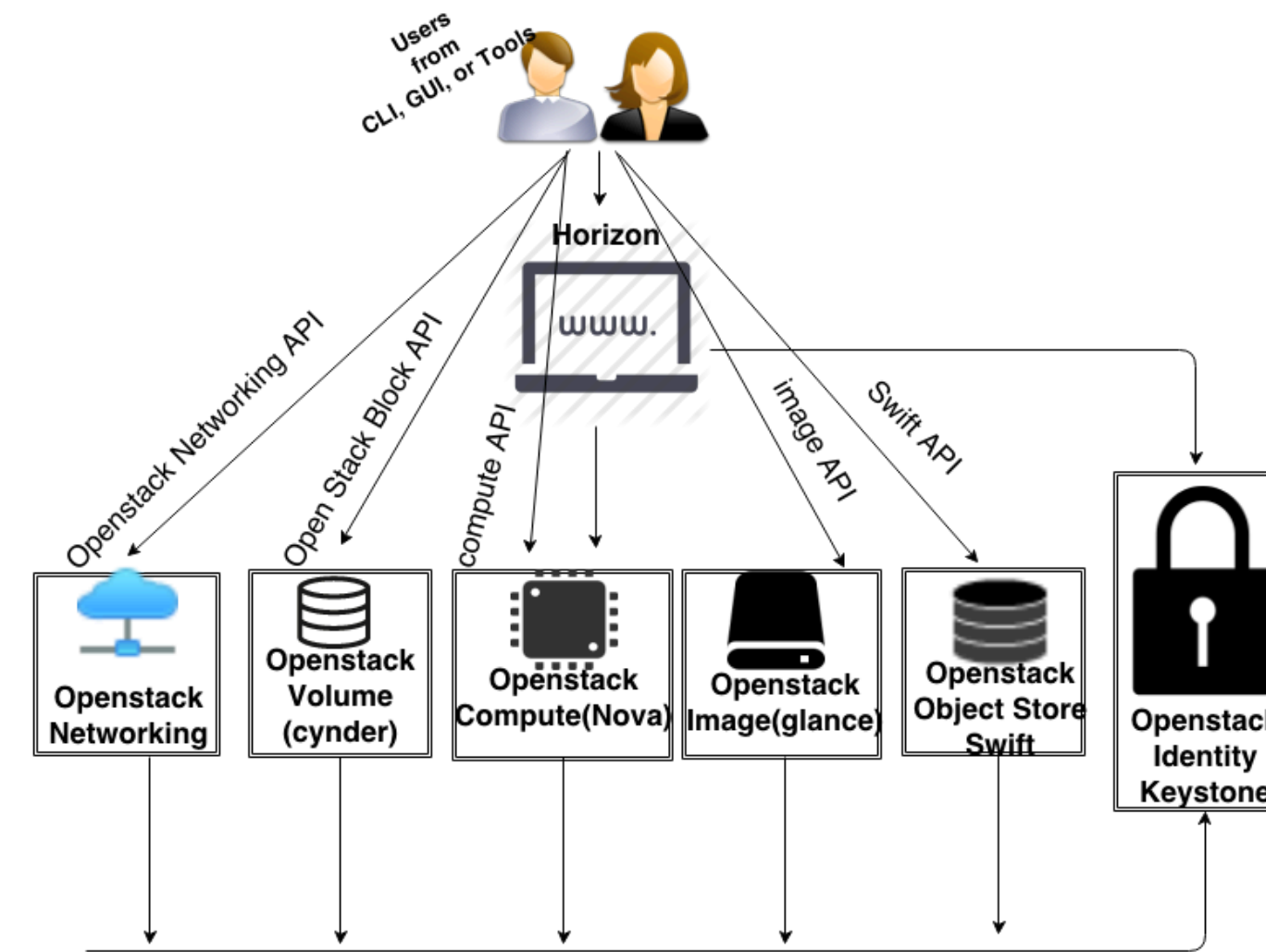


OPENSTACK ARCHITECTURE

OpenStack is a modular IaaS platform. Each service plays a key role:

- Nova – Manages virtual machines by controlling hypervisors like KVM.
- Neutron – Handles networking: virtual networks, subnets, routers, and security groups.
- Glance – Stores and manages VM images like Ubuntu or CirrOS.
- Cinder & Swift –
 - Cinder provides block storage (like a virtual hard drive).
 - Swift offers object storage for unstructured data (like backups).
- Horizon – A web dashboard to manage all services via a user-friendly interface.

These components communicate through APIs, with identity handled by Keystone (not shown), forming a complete cloud infrastructure.



VIRTUAL MACHINE DEPLOYMENT

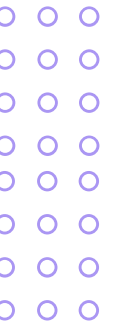


- Used Nova to create VMs
- Assigned floating IPs for external access
- Installed Apache/Nginx via SSH
- Configured security groups





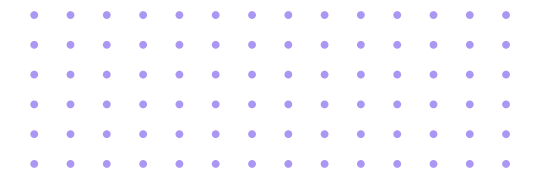
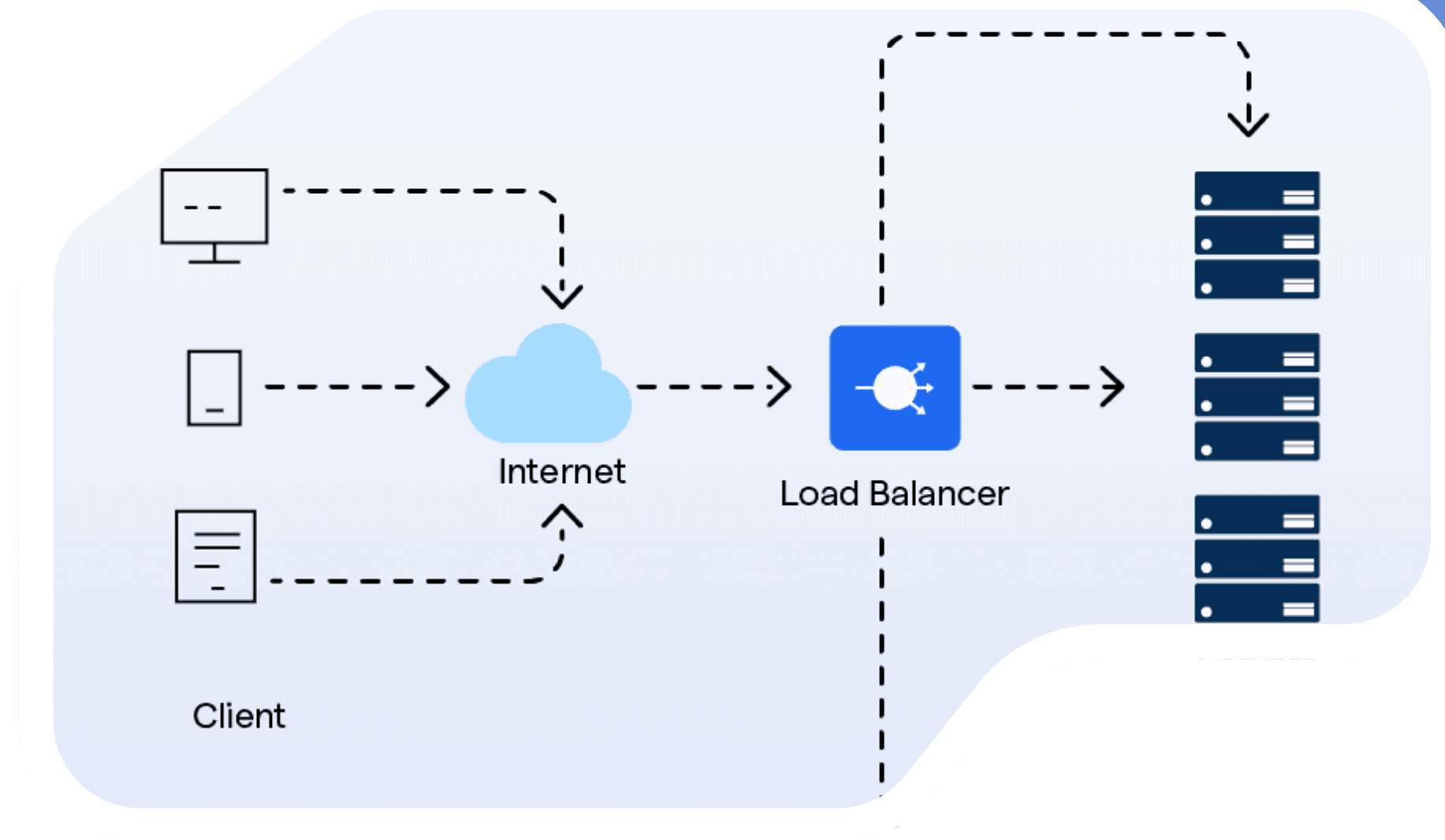
STORAGE AND NETWORKING



- Cinder: persistent block storage
- Swift: object storage for files
- Neutron: setup of private/public networks
- VMs linked via routers

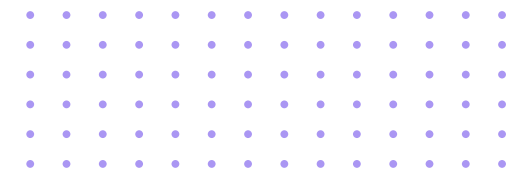
LOAD BALANCING & AUTO-SCALING

- Octavia (LBaaS): traffic to multiple VMs
- Heat: auto-scale based on load (CPU)
- Benefits: High availability, elasticity

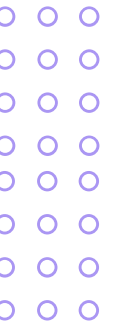


CHALLENGES AND SOLUTIONS

- Errors: missing packages, service crashes
- Fixes: update system, correct IPs, log debugging
- Lessons: prepare environment, monitor logs, test step by step



CONCLUSION



- Full-cycle cloud deployment experience
- Hands-on with VMs, storage, networking
- Improved problem-solving and teamwork
- Ready for DevOps & cloud roles



